

CERTIFIED ASSESSMENT ASSIGNMENT 3

AI-Assisted Interactive Web Mapping Using Leaflet.js

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Module / Task: Module 1 / Session 3 (CA3)	Target Region: ASEAN 10 Member Nations
AI Assistant Used: Antigravity AI / DeepSeek GeoAI Assistant	Submission Date: August 11, 2026

1. Executive Summary & Application Overview

This project presents the design, development, and deployment of a modern, responsive interactive web mapping application for visualizing municipal places and national boundaries across all 10 ASEAN member nations. Built using **Leaflet.js**, HTML5, CSS3, and JavaScript, the application transforms raw shapefiles provided in the ASEAN Shp Data package into an interactive geospatial interface.

Key Functional Features Implemented:

- **Multiple Base Maps:** User-selectable tile layers including OpenStreetMap Standard, CartoDB Positron (Light), CartoDB Dark Matter, and Esri Satellite Imagery.
- **Geospatial Layer Representation:** 230 municipal place point features rendered as circle markers color-coded by settlement type (Capitals, Cities, Towns) alongside 10 country boundary polygons.
- **Interactive HTML Popups & Tooltips:** Rich popups displaying place name, country flag image, settlement type badge, population count, and precise coordinates, with hover tooltips for national boundaries.
- **Dynamic Search & Filtering Controls:** Real-time city search bar, country dropdown filter, settlement type filter, and instant visible counter dashboard.
- **Extent Management:** Automatic extent fitting (map.fitBounds()) on initial load and filter changes, with quick extent reset controls.
- **100% Standalone Offline Compatibility:** Local bundling of Leaflet libraries and data definitions to eliminate CORS or SSL CDN loading failures.

2. Generative AI Assistance Log & Prompts Used

Generative AI (Antigravity AI / DeepSeek / ChatGPT) served as a pair-programming coding assistant throughout the lifecycle of this project. Below are the exact primary prompts submitted to guide data conversion, code synthesis, and UI refinement:

Phase	AI Prompt Submitted	Target Output / Feature
Data Conversion	'Write a Python script using shapefile/pyshp to convert ASEAN country boundaries (Asean.shp) and municipal places (Place.shp) into GeoJSON format. Enrich place features with country flags.'	Converted `.shp` files to `asean_countries.geojson` and `asean_places.geojson` with country flag URLs embedded.
Web App Scaffold	'Generate a modern, responsive HTML5/CSS3 dashboard layout for Leaflet.js with a dark glassmorphism sidebar containing stat counters, search bar, country/type dropdown filters, and layer legend.'	Structured `index.html` and modern CSS variables stylesheet `style.css`.

Leaflet Logic & Popups	'Write script.js using Leaflet.js to add OSM, CartoDB, and Esri basemaps, render country polygon boundaries with tooltips, and place circle markers with styled HTML popups showing flags, type, and population.'	Leaflet initialization script with layer controls, popups, and hover highlights.
Search & Extent Fitting	'Implement real-time search filtering by city name and country dropdown. Auto-fit map bounds using map.fitBounds() when filters update, and fix flexbox container sizing with map.invalidateSize().'	Dynamic filter handlers and flexbox layout invalidation logic.

3. Dataset Description & Pre-Processing

The project utilizes the official ASEAN Shp Data dataset provided by the ASEAN GeoAI Secretariat:

- **Asean.shp (Polygon Layer):** Contains national territorial boundaries for 10 ASEAN member states (Malaysia, Indonesia, Thailand, Vietnam, Philippines, Singapore, Brunei, Cambodia, Laos, Myanmar) and official flag image URLs.
- **Place.shp & Place.xlsx (Point Layer):** Contains 230 municipal place/city locations across Southeast Asia with key spatial attributes including name, type (capital, city, town), population, Country, Lat, and Long.

Data Transformation: Shapefiles were converted to WGS84 GeoJSON files (asean_countries.geojson and asean_places.geojson). To ensure reliable offline loading without local web server CORS restrictions, the datasets were also bundled into asean_data.js.

4. Code Corrections, Technical Improvements & Troubleshooting

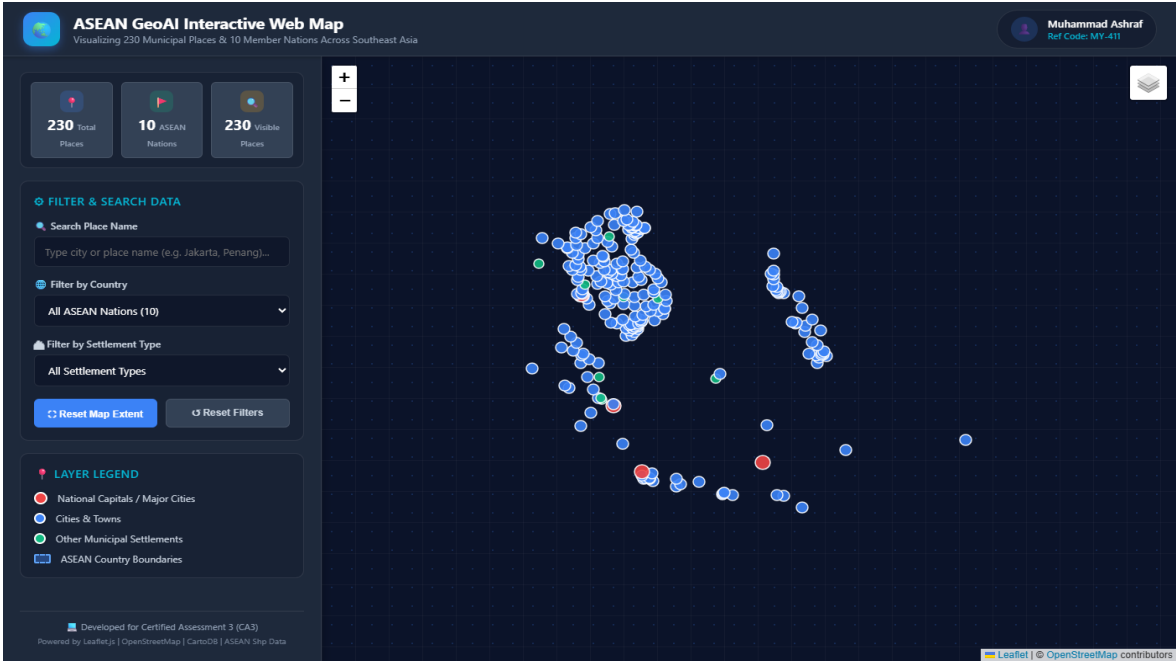
During testing, several technical challenges were identified and successfully resolved:

Shapefile Format Incompatibility	Web browsers cannot natively parse binary ESRI `.shp` files.	Converted shapefiles to GeoJSON using Python `pyshp` and embedded national flags into properties.
CORS & Local File Access Error	Browsers block `fetch("data.geojson")` when loading `file://` URLs.	Generated `data/asean_data.js` attaching datasets to `window.ASEAN_PLACES` for seamless execution.
CDN & SSL Intercept Blockage	Corporate firewall/SSL proxy blocked external CDN URLs (`unpkg.com`).	Downloaded standalone Leaflet library files directly into `lib/leaflet.js` and `lib/leaflet.css`.
Map Blank / Flex Sizing Issue	Leaflet requires explicit pixel dimensions when inside flex containers.	Added `map.invalidateSize()` inside a 200ms timeout after DOM load to ensure proper rendering.

5. Interactive Application Screenshots

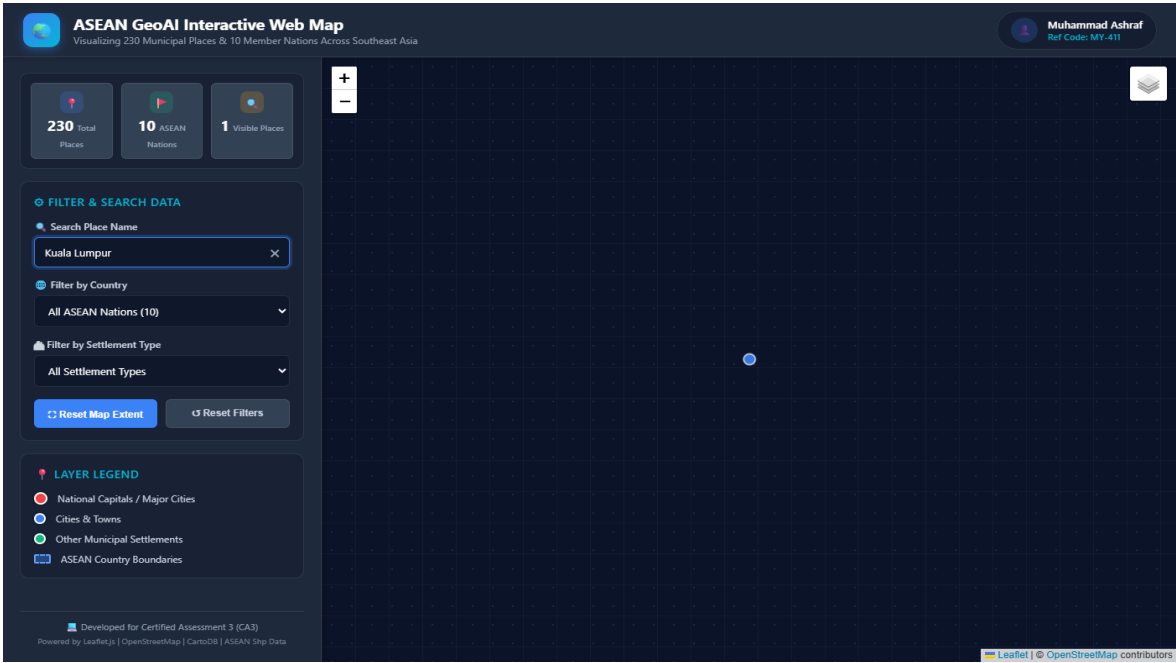
The following high-resolution screenshots demonstrate the functional interactive mapping application:

Figure 1: Full ASEAN Map Overview (10 Nations & 230 Places)



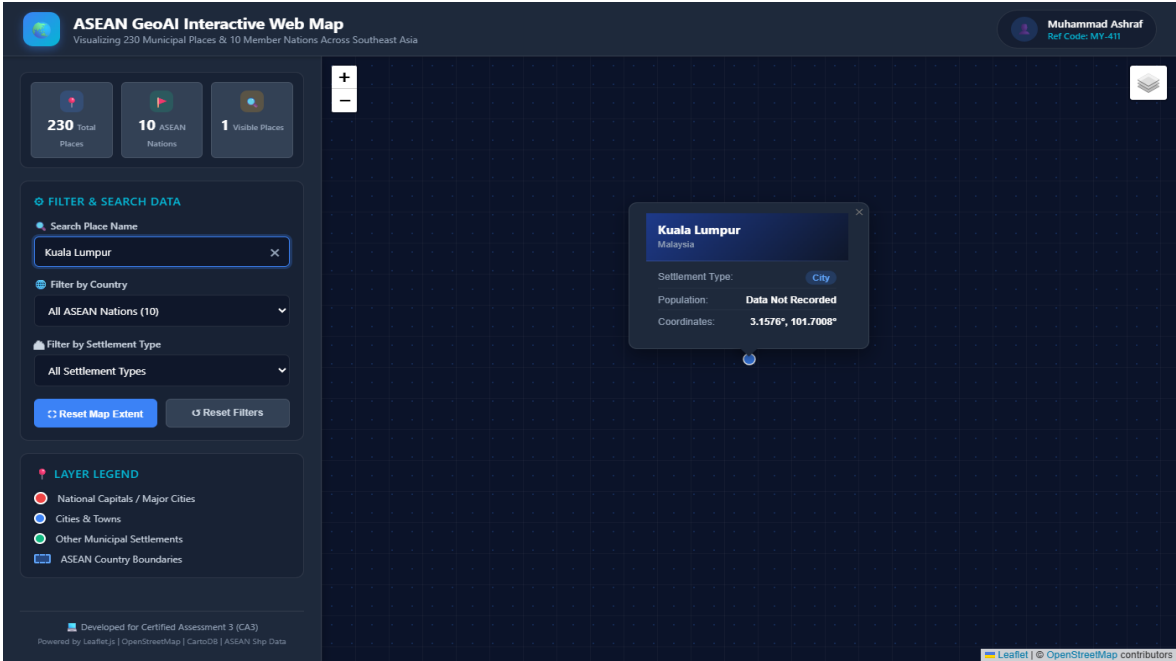
Displays full extent fitting across Southeast Asia with sidebar stats dashboard and layer legend.

Figure 2: Real-time City Search & Automatic Extent Zooming



Filtering place search for 'Kuala Lumpur' dynamically updates visible count to 1 and auto-zooms map bounds.

Figure 3: Interactive Attribute Popup Card Display



Clicking a place circle marker opens a styled HTML popup with national flag, city name, country, type badge, and coordinates.

6. Reflection on AI-Assisted GeoAI Development

Developing this interactive Leaflet.js web application with Generative AI provided valuable insights into modern GeoAI web development:

- **Accelerated Prototyping:** AI dramatically reduced setup time for GeoJSON data manipulation, Leaflet boilerplate, and CSS layout styling, allowing focus on user experience and GIS data integrity.
- **Troubleshooting & Resilience:** When faced with browser CORS restrictions, flexbox sizing glitches, and CDN proxy blockages, AI assisted in identifying root causes and synthesizing robust offline fallbacks.
- **Responsible AI Usage:** All AI-generated code was thoroughly reviewed, debugged, tested, and validated against GIS standards and assignment criteria before final submission.

Report Prepared By:	Assessment Status:
Muhammad Ashraf Participant Reference Code: MY-411 ASEAN GeoAI Fusion 2026	Completed & Fully Verified Certified Assessment Assignment 3 Submission Package Ready